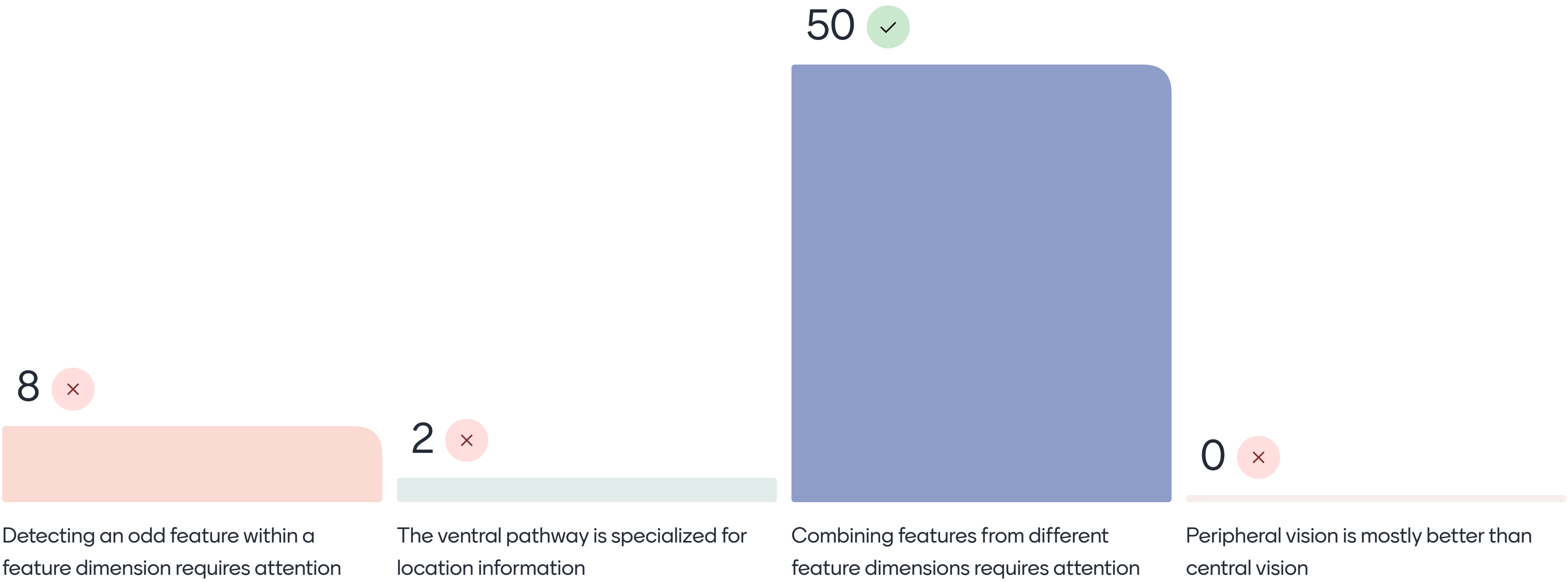
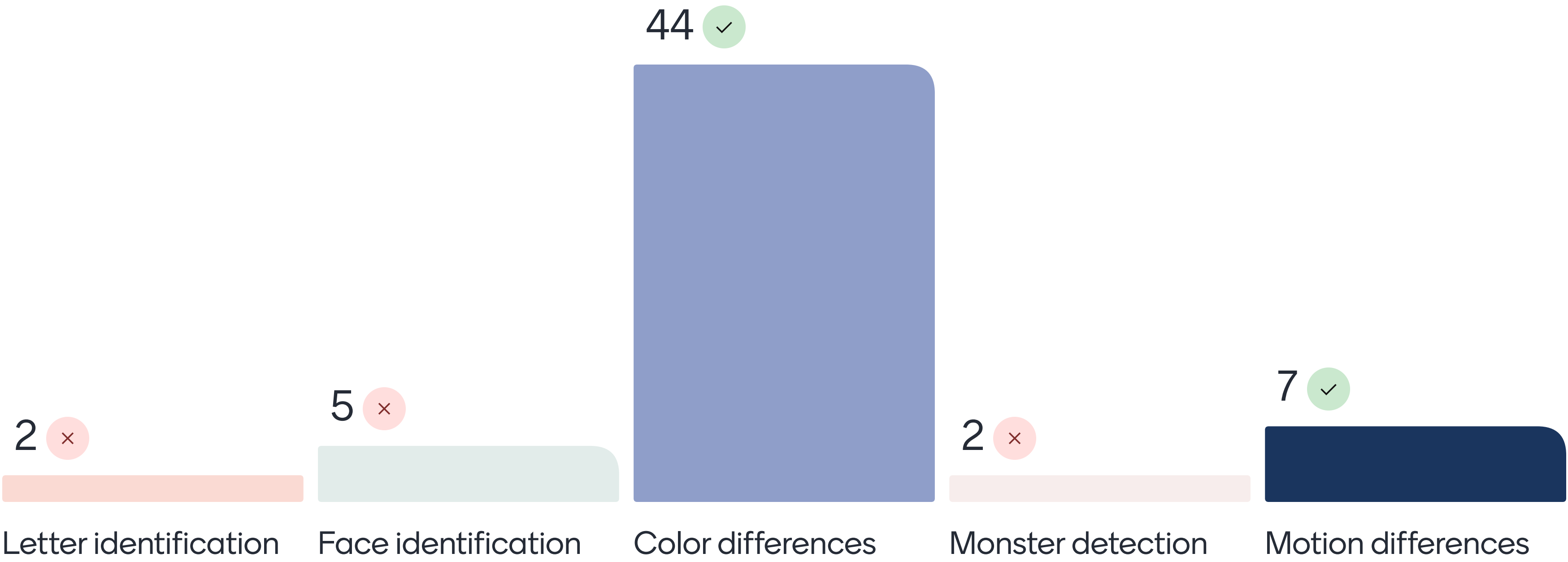


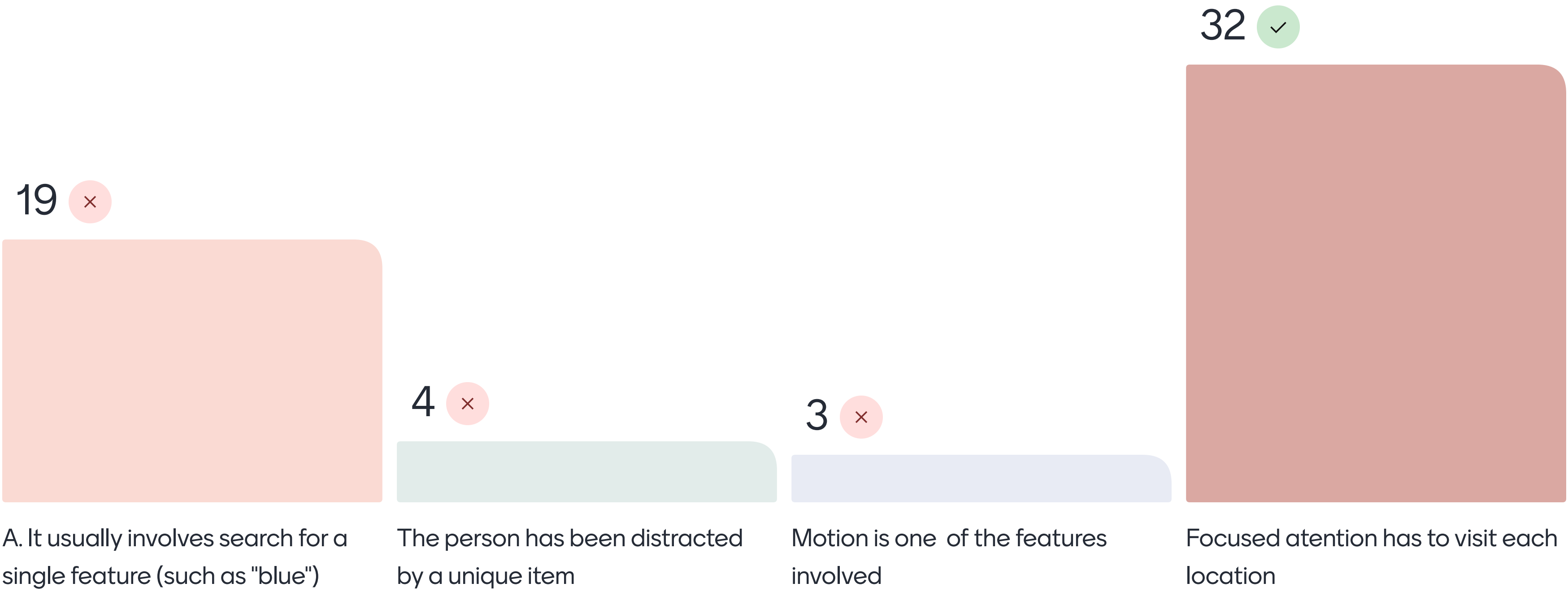
# Which is thought to be true?



# Simultaneous parallel processing across the visual field occurs for:



According to Treisman's Feature Integration Theory, when search is serial, what is also true?



# Describe one of the main points:

Wheres Wally

Where's Wally

Where's Wally

Where's Wally

Where's Wally

Where's Wally

Where's Wally

Where's Wally

# Describe one of the main points:

Where's Wally

Where's Wally

Where's Wally

Where's Wally

serial processing - odd  
pair of features

Where's Wally

Where's Wally

It takes a lot more time for  
different feature values to be  
processed at the same time,  
making it harder to spot 'the  
odd one out' in such situations





# Describe one of the main points:

feature selection and location selection are separate - you cannot look for red in one area and blue in another

Where's Wally

we process what and where in different parts of our brain

with parallel the amount of distractors doesn't matter

Where's Wally

Finding a single odd feature is easy and instantaneous, but combining features such as orientation and colour is slower and requires precise attention

Where's Wally

Where's Wally

# Describe one of the main points:

Where's Wally

When searching for a single feature, the process is parallel, but when searching for a combination of features, the process becomes serial due to constraints on the bottle neck

Where's Wally

you can't combine selecting a simple feature and selecting a stimuli in a specific region

feature integration theory of triesman, response time will go up with the number of objects increasing

Where's Wally

it is very hard to find woldo

Where's Wally



# Describe one of the main points:

no

Where's Wally

Combination of features  
requires attending to the  
location

combination of features  
takes much longer to  
identify than when you are  
focusing on a single feature.

FIT = pre-attentive parallel  
feature maps + serial,  
attention-based binding  
stage

Where's Wally

Where's Wally

Feature selection is global  
and allows your attention to  
go straight to certain  
targets



# Describe one of the main points:

Where's Wally

Bringing together the What and Where pathways is difficult to achieve

Where's Wally

focusing on one feature (red in a stimulus of majorly blue circles with one red circle) allows brain to inhibit similar neighbouring features (circles and the colour blue)

can parallel process for odd one out within a single feature domain, but finding the unique combination requires a serial search

Where's Wally

finding wally = finding my friend on a music festival lol rip

Where's Wally



# Describe one of the main points:

our peripheral vision is not good

It doesn't matter how many additional features there are, you see everything in parallel?

Where's Wally

Parallel processing always occurs at the same speed, but for serial processing as the amount of distractors increases more time/effort is required

Where's Wally

Feature selection is global because that's how the what-where pathway system works

- Feature attention is **global**, location attention is **local**, the two can't combine - Reflects the **dorsal/ventral** pathway split

Where's Wally

# Describe one of the main points:

when parallel processing (unique feature) amount of distractors doesn't inc the amount of time. when searching for a combination / serial processing - amount of distractors increases time (steep slope)

When looking at a single dimension (colour, motion, orientation) search time doesn't depend on how many distractors there are.

Where's Wally

while we are good at spotting the out one out in one dimension (eg odd colour/orientation etc) it is much more effortful to combine features (eg orientation and colour) to find the odd one out

Where's Wally

There is nuance to parallel vs. serial. We are only able to (select?) parallelly when there is 1 feature. combining features we need our serial spotlight!

Where's Wally

We can't combine feature dimensions and values

# Describe one of the main points:

Where's Wally

Where's Wally

During parallel processing, it doesn't matter if you increase the number of items, the time taken to find the odd one out will be the same.

Where's Wally

we can't process different features simultaneously

Where's Wally

Where's Wally

Feature combination is a slow process

# Describe one of the main points:

parallel feature processing is heavily reliant on the visual cortex

In the example of the odd ball. The odd colour releases inhibition which in turns allows us to spot it right away

Human attention involve different pathways of the brain that helps us to focus on different things or figure out the unique one.

Treisman's Feature Integration Theory theorized that different features are processed simultaneously and are combined later.

Having distractors also doesn't work on a sniffer dog's sense of smell (parallel olfactory feature processing?)

wally missed his assessment so we need to find him -- From FBI

When having to look for a combination of different features, you must actively attend to each area.

identifying feature combination is usually slow



# • Distinguish the ways a stimulus can be selected

- Contrast *bottom-up (salience-driven)* and *top-down (goal-directed)* attentional control.
- Identify the types of top-down selection: spatial/location and feature selection.

# • Explain why feature selection is global (CET)

## Things I didn't understand

- State that attending to a feature (e.g., red) enhances that feature across the *entire* visual field, not just a restricted region.
- Explain why you cannot simultaneously attend to red on the left and blue on the right.

Honestly everything was pretty straight forward

The FIT of Treisman, model is confusing

na

Treisman thing? i feel like we didn't actually go through the levels and meanings of that diagram and theory

I feel like I did but I'd be happy to do it again one-on-one if you email me

Just a question: If for example, two colours are the same but have different slightly different hues/shades, do we still find the odd one out at the same rate?

Is serial processing requiring effort the same for all values? (Is colour easier than direction?)

That's difficult to compare because it's comparing apples and oranges, how much of a color differences, with what spacing, is equivalent to a particular motion direction set? (See previous answer to get a greater sense of the complications)

-

Caleb owens

Good question, if the colors are too close in hue, brightness, or saturation, parallel search breaks down and becomes slow.

- Explain how *illusory conjunctions* arise when features are incorrectly bound without focused attention.
- Apply FIT to predict whether a given visual search will be easy (parallel) or difficult (serial).



- Distinguish the ways a stimulus can be selected
  - Contrast *bottom-up (salience-driven)* and *top-down (goal-directed)* attentional control.
  - Identify the types of top-down selection: spatial/location and feature selection.

# Things I didn't understand

- Explain why feature selection is global (C12)
  - State that attending to a feature (e.g., red) enhances that feature across the *entire* visual field, not just a restricted region.
  - Explain why you cannot simultaneously attend to red on the left and blue on the right.

Where's Wally

most of it

The FIT diagram

Why the what-where pathway design means only one feature dimension can be processed at a time

Good question, that was actually specific to explaining why you can't do one feature in some locations and another in others. In other words, it was about the inability to pair *\*locations\** (where) with a *\*feature\** (what)

there are types of top-down selection (spatial and feature selection) but are there types of bottom-up selection that we should know

So motion and colour are the only features that can be searched for parallel? I thought we just need to have ONE of any feature not just these.

caleb owens

- Explain how *illusory conjunctions* arise when features are incorrectly bound without focused attention.
- Apply FIT to predict whether a given visual search will be easy (parallel) or difficult (serial).

# Thing(s) I found interesting / other comments

All the weird characters  
in where's wally lol

na

-

Can you improve serial  
processing with training?  
*See Chapter 11 (not covered this year)*

Where's Wally

All of it

the active  
demonstrations

motion detection when  
colour switches and is fast is  
very difficult/impossible





# Thing(s) I found interesting / other comments

the red and blue dots  
example was cool

Glad it worked for you!

i think people got it wrong  
because choice A would  
have been correct if it was  
talking about parallel not  
serial

Good idea - so not paying enough  
attention (less likely in an exam)  
or not yet having internalised the  
relevant distinctions

parallel feature  
processing concept